

SENSORS AND THEIR FUNCTIONS

Sensor	Function
Engine speed sensor	Monitors spinning speed of crankshaft
Fuel temperature sensor	Ensures right amount of fuel is injected to keep motion smooth
Spark knock sensor	Ensures fuel is burned correctly
Voltage sensor	Manages car speed and ensures speed is controllable
MAP sensor	Measures manifold pressure inside and outside
Oxygen sensor	Measures unburden oxygen presented in exhaust pipe
MAF sensor	Calculates density and volume of air taken in by engine
Air-fuel ratio meter	Monitors correct air-fuel ratio of engine
Crank position sensor	Monitors piston's top-dead centre (TDC) position in engine
Cam position sensor	Monitors position of valves in engine
Throttle position sensor	Monitors position of throttle in engine
Knock sensor	Detects engine knocking due to timing advance
Engine coolant temperature sensor	Measures engine temperature



Fig. 2: Typical ECU (Credit: Wikipedia)

air control valve and exhaust gas re-circulation (EGR) valve. Then, it extracts the best possible engine performance while keeping emissions as low as possible.

Engine fault diagnosis. ECU collects signals from various sensors, including faulty ones, and stores these in its memory. Sensors diagnose these faults either by reading ECU memory directly or engine diagnostic equipment supplied by the vehicle manufacturer.

Modern luxury cars contain hundreds of ECUs, but cheaper and smaller cars only a handful. The number of ECUs goes up with ever-increasing features.

Depending on the vehicle make and model, the ECU(s) can be found beneath the wiper, under the bonnet in engine bay, passenger front footwell under the carpet or near the glove compartment.

Some common vehicle sensors include ambient light, battery current, differential oil temperature, door open warning, anti-lock brak-

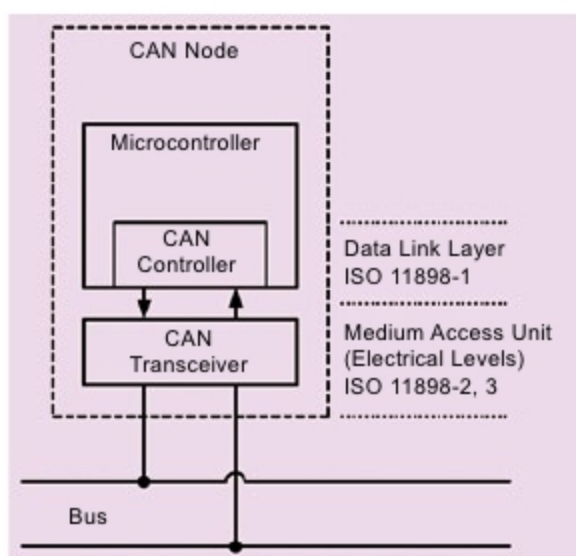


Fig. 3: CAN bus node (Credit: Wikipedia)

ing system (ABS), auto door lock position, battery temperature, brake power booster, camshaft position, crankshaft position, cylinder head temperature, diesel emissions fluid temperature, fuel cutoff, fuel temperature, headlight level, humidity, hybrid battery voltage, hybrid circuit breaker, ignition pass-lock, manifold absolute pressure (MAP), mass air flow (MAF), oil level, oxygen, power steering fluid level, speed, steering angle, temperature, throttle position, transmission oil pressure and windshield washer level.

As a car owner, you do not have to know the functions of all sensors used in the vehicle, but knowing the basics can be helpful in case your car breaks down or something else goes wrong. Getting a replacement sensor demands attention to genuineness, quality and performance of the product. To understand these

functions, a list of some popular sensors used in modern vehicles are listed in the table.

Sensors in autonomous vehicles

With technological advancements, a car can now drive itself. Navigating a car from origin to destination is possible with autonomous vehicle technology (that is, without human driver), including avoiding road hazards and responding to traffic conditions. Modern sensors and technology can even help a driverless car to drive at high speeds on open road.

Autonomous cars use many sensors including radars and cameras. Lidar is the primary sensor used in most driverless cars. It helps in sensing the world around and bouncing laser light off nearby objects to create 3D maps of the surroundings. Lidar does not detect objects; it profiles these by illuminating these and analysing the path of reflected light. It uses emitted light and yields high-resolution images. It is not affected much by the intensity of light at any time (day or night) and, hence, the result is extremely accurate.

Autonomous vehicles have been around for almost a decade, but lidar is relatively new and expensive. Making it work on practical autonomous cars is not easy. The cars must be robust and reliable. These should possess some form of artificial intelligence (AI) to be able to take care of rough paths, collisions, obstacles, potholes, traffic, etc. Although, a lot of improvement is still required, lidar has solved many problems in autonomous cars. Therefore it is an indispensable component for fully-autonomous cars.

Autonomous vehicles demand complex integration of sophisticated algorithms running on powerful processors, making critical decisions based on real-time data coming from a diverse and complex array of sensors. The vehicles need good and reliable sensors including GPS, cameras, MEMS-based gyroscopes and accelerometers. Although, the race for autonomous cars has begun, it has some limitations due to the limi-